CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cell biologist adds a poison that blocks the mitochondria. The process first affected is: 1

(a) energy release **(b)** osmosis
(c) diffusion **(d)** protein synthesis

2. Cell sap is found in the: 1

(a) mitochondrion **(b)** Golgi **(c)** vacuole **(d)** nucleus

3. Assertion (A): Muscle cells contain a large number of mitochondria. Reason (R): Muscle cells need a lot of energy for contraction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The number of cells after 4 successive mitotic divisions of one cell is: 1

(a) 4 **(b)** 8 **(c)** 32 **(d)** 16

5. A cell biologist adds a poison that blocks ribosomes. The cell will fail to make: 1

(a) proteins **(b)** water **(c)** ATP **(d)** lipids

6. Chromoplasts are found in: 1

(a) bones **(b)** bacteria
(c) roots **(d)** petals and ripe fruits

7. Assertion (A): Lysosomes are called suicide bags. Reason (R): They contain digestive enzymes that may digest the cell itself when damaged. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. The number of gametes formed from 10 primary spermatocytes is: 1

(a) 40 **(b)** 10 **(c)** 20 **(d)** 5

9. Cells of a leaf show green colour, but cells of a root do not because: 1

(a) roots have no nucleus **(b)** roots have no wall
(c) root cells lack chloroplasts **(d)** leaves have more vacuoles

10. Leucoplasts are found in: 1

(a) potato tubers **(b)** RBC **(c)** petals **(d)** leaves

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How does a cell obtain energy from glucose? Name the organelle and the process. 2

12. Why are the animal cells flexible but plant cells rigid? 2

13. Explain the role of the nucleolus in protein synthesis. 2

14. What would happen if all the lysosomes burst in a cell? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cell has 2n = 8. Describe the number of chromosomes at the end of mitosis and at the end of meiosis I and II. 3

16. Explain how the plant cell, though surrounded by a rigid wall, manages to transport substances through plasmodesmata. 3

17. How does the cell cycle ensure that the DNA is copied accurately before mitosis? 3

OR

Explain why a mature plant cell has a large vacuole but a meristematic cell has many small ones. 3

18. The diameter of a cell is 50 micrometres and of a bacterium 2 micrometres. Compare the volumes and the SA/V ratio (spheres). 3

OR

A cell lacking the Golgi apparatus cannot secrete some proteins. Explain the effect with the example of a gland cell. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the sequential events in the production of a secretory protein such as insulin, naming each organelle and process, and the energy source. 5

OR

Compare diffusion in a cell with the rate of transport in a big organism, using the SA/V ratio of cubes of sides 1, 2 and 4 cm. Calculate and explain the need of special organs for exchange in larger organisms. 5

20. A plant cell in a hypotonic solution absorbs water and becomes turgid. Explain the changes in water potential, the pressure on the wall and the way the equilibrium is attained. Also state the effect of cell wall removal. 5

OR

Plan a project on the effect of temperature on the permeability of the cell membrane using beetroot discs. State the method, controls, measurements, expected observations and the explanation. 5

END OF QUESTION PAPER — Total: 40 Marks